

4. The two integrals below should be done using trig substitution. Tell the substitution that should be made. (5 pts each)

(a)  $\int \frac{\sqrt{25-9x^2}}{5x^2} dx$

(b)  ~~$\int \sqrt{x^2-8x+41} dx$~~

On the remaining problems, evaluate the integral.

5.  $\int x \sec^2 x dx$  IBP  $uv - \int v du$

(10 pts)

$$\boxed{\begin{array}{l} u = x \quad dv = \sec^2 x dx \\ du = dx \quad v = \tan x \end{array}}$$

$$\int \frac{\sin x}{\cos^2 x} dx$$

$$= x \tan x - \int \tan x dx$$

$$\boxed{= x \tan x - \ln |\cos x| + C}$$

6.  $\int \frac{2x-5}{x^2+4} dx = \int \frac{2x}{x^2+4} dx - \int \frac{5}{x^2+4} dx$

(10 pts)

$$\begin{array}{l} u = x^2 + 4 \\ du = 2x dx \end{array} = \int \frac{du}{u} - \frac{5}{2} \arctan\left(\frac{x}{2}\right) + C$$

$$= \ln |u| \quad "$$

$$= \ln(x^2 + 4) - \frac{5}{2} \arctan\left(\frac{x}{2}\right) + C$$

Name: \_\_\_\_\_

7.  $\int 3x^2 e^{2x} dx$

(10 pts)

$$\begin{array}{rcl} + 3x^2 & \searrow & e^{2x} \\ - 6x & \searrow & \frac{1}{2} e^{2x} \\ + 6 & \searrow & \frac{1}{4} e^{2x} \\ 0 & \searrow & \frac{1}{8} e^{2x} \end{array}$$

$$3x^2 \cdot \frac{1}{2} e^{2x} - 6x \cdot \frac{1}{4} e^{2x} + \frac{6}{8} e^{2x} + C$$

$$\left( \frac{3}{2} x^2 - \frac{3}{2} x + \frac{3}{4} \right) e^{2x} + C$$

$$\frac{s^2}{c^2} + \frac{c^2}{c^2} = \frac{1}{c^2}$$

8.  $\int \sec^5 x \tan^3 x dx$

$$\tan^2 x + 1 = \sec^2 x$$

(10 pts)

$$\int \sec^3 x \tan^3 x \frac{\sec^2 x dx}{du} \quad \text{or} \quad \int \frac{\sec^4 x}{u^4} \underbrace{\tan^2 x}_{u^2-1} \underbrace{\sec x \tan x dx}_{du}$$

BAD

$$u = \sec x$$

$$du = \sec x \tan x dx$$

$$\int u^4 (u^2 - 1) du = \int u^6 - u^4 du = \frac{1}{7} u^7 - \frac{1}{5} u^5 + C$$

$$= \boxed{\frac{1}{7} \sec^7 x - \frac{1}{5} \sec^5 x + C}$$

$$\sin^2 x = \frac{1}{2} [1 - \cos 2x]$$

$$9. \int_0^{\pi} 4 \sin^2 x \, dx = 4 \cdot \frac{1}{2} \int_0^{\pi} 1 - \cos 2x \, dx$$

(10 pts)

$$= 2 \left[ x - \frac{1}{2} \sin 2x \right]_0^{\pi} = 2 \left[ \pi - \frac{1}{2} \sin 2\pi \right] - 2 \left[ 0 - \frac{1}{2} \sin 0 \right]$$

$$= \boxed{2\pi}$$

$$10. \int \frac{10x^2 - 15x + 2}{(x-2)x^2} \, dx$$

(10 pts)

$$(x-2)x^2 \left[ \frac{10x^2 - 15x + 2}{(x-2)x^2} = \frac{A}{x-2} + \frac{B}{x} + \frac{C}{x^2} \right]$$

$$10x^2 - 15x + 2 = \underline{Ax^2} + \underline{Bx^2 - 2Bx} + C(x-2)$$

$$\underline{x=0}$$

$$2 = C(-2)$$

$$\boxed{-1 = C}$$

$$\underline{x=2}$$

$$40 - 30 + 2 = A \cdot 4 + 0 + 0$$

$$12 = 4A$$

$$\boxed{3 = A}$$

$$\frac{x^2}{x^2}$$

$$10 = A + B$$

$$7 = B$$

$$\int \frac{3}{x-2} + \frac{7}{x} - \frac{1}{x^2} \, dx = \boxed{3 \ln|x-2| + 7 \ln|x| + \frac{1}{x} + C}$$

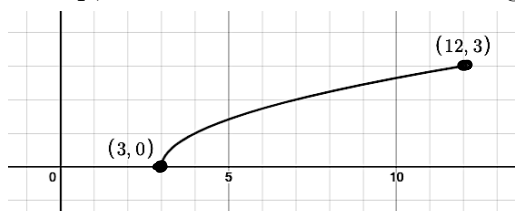
$$x^{-2} \rightarrow \frac{x^{-1}}{-1}$$

11. Find the smiley.

(2 pts)



1. A segment of the curve  $x = y^2 + 3$  is shown below.



$$\frac{dx}{dy} = x' = 2y$$

$$dS = \sqrt{1 + \left[\frac{dx}{dy}\right]^2} dy$$

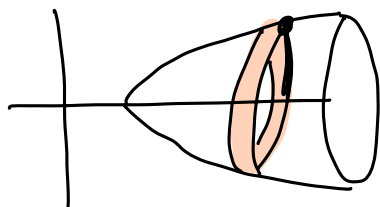
(a) Set up, but **do not evaluate** an integral for the length of the segment.

(6 pts)

$$\int_0^3 \sqrt{1 + 4y^2} dy$$

(b) Find the surface area of the surface obtained by rotating the segment around the  $x$ -axis and give the exact area.

(10 pts)



$$A_{\text{band}} = \frac{2\pi(\text{rad}) dS}{\text{Circumference width}}$$

$$\text{radius} = y$$

$$S.A. = 2\pi \int_0^3 y \sqrt{1 + 4y^2} dy$$

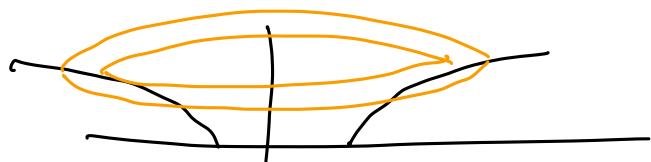
$$u = 1 + 4y^2 \quad du = 8y dy$$

$$2\pi \int_1^{37} \frac{1}{8} (u)^{1/2} du$$

$$= \frac{\pi}{4} \int_1^{37} u^{1/2} du = \frac{\pi}{4} \left[ \frac{2}{3} u^{3/2} \right]_1^{37} = \boxed{\frac{\pi}{6} [37^{3/2} - 1]}$$

(c) Set up, but **do not evaluate** an integral for the surface area obtained by rotating the segment around the  $y$ -axis

(6 pts)



$$\text{radius} = x$$

$$S.A. = 2\pi \int_0^3 x \sqrt{1 + 4y^2} dy$$

$$= \boxed{2\pi \int_0^3 (y^2 + 3) \sqrt{1 + 4y^2} dy}$$

2. Classify these improper integrals as convergent (finite) or divergent (infinite). (10 pts)

Write C or D in each blank

D (a)  $\int_2^\infty \frac{1}{\sqrt{x}} dx$   
 $\sum \frac{1}{\sqrt{n}}$

C (b)  $\int_2^\infty \frac{1}{x^{1.5}} dx$   
 $\sum \frac{1}{n^{1.5}}$

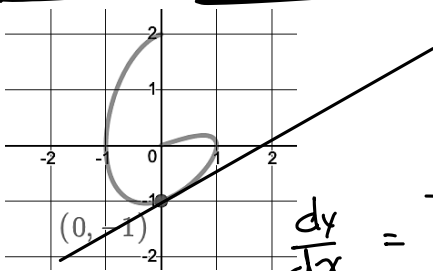
D (c)  $\int_2^\infty \frac{1}{x} dx$   
 $\sum \frac{1}{n}$

C (d)  $\int_4^\infty \frac{1}{x^2} dx$

D (e)  $\int_2^\infty \left( \frac{1}{x^3} + \frac{1}{\sqrt{x}} \right) dx$   
 $\sum \frac{1}{n^3} \quad \sum \frac{1}{\sqrt{n}}$

3. Find the equation of the line that is tangent to this parametric curve at the point with  $t = 1$  (which is labeled on the graph). (10 pts)

$x = \sin(\pi t)$     $y = t \cos(\pi t)$



$$m = \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

$$\frac{dy}{dx} = \frac{-t \sin(\pi t) \cdot \pi + \cos(\pi t) \cdot 1}{\cos(\pi t) \pi} =$$

$$t=1 \Rightarrow \frac{-\overset{0}{\sin(\pi)} \cdot \pi + \overset{(-1)}{\cos(\pi)} \cdot 1}{\underset{(-1)}{\cos(\pi)} \pi} = \frac{1}{\pi} = m$$

$$\boxed{y = \frac{1}{\pi} x - 1}$$

4. The position of a moving particle is given by the parametric equations below. Give your answers in interval notation: For which value(s) of  $t$  is the particle ... (10 pts)

$x = \frac{1}{4}t^4 - \frac{9}{2}t^2$     $y = \frac{3}{2}t^2 - 7t + 1$

(a) moving to the right?

(b) moving down?

when  $\frac{dx}{dt} > 0$

when  $\frac{dy}{dt} < 0$

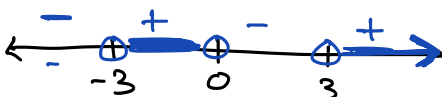
$$\frac{dx}{dt} = t^3 - 9t = 0$$

$$t(t^2 - 9) = 0$$

$$t = 0 \quad t^2 = 9$$

$$t = 0, t = 3, t = -3$$

$\frac{dx}{dt}$

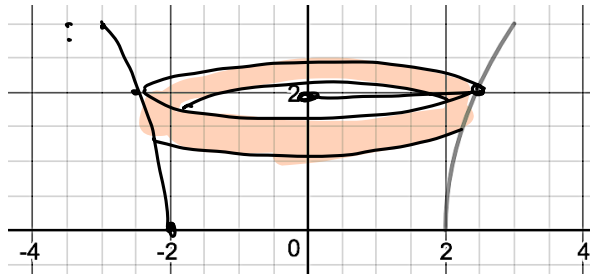


$$(-3, 0) \cup (3, \infty)$$

$$dS = \sqrt{\left(\frac{dy}{dt}\right)^2 + \left(\frac{dx}{dt}\right)^2} dt$$

5. For this parametric curve, find the following things. Do not leave any  $x$ 's or  $y$ 's in your answers.

$$\begin{aligned} x &= \underline{3t} \\ y &= \underline{t^2 + 2} \\ 0 &\leq t \leq \underline{1} \end{aligned}$$



- (a) Find  $dS$ .

(2 pts)

$$dS = \sqrt{(3)^2 + (2t)^2} = \sqrt{9 + 4t^2} dt$$

- (b) Set up **but do not evaluate** an integral for the arc length.

(5 pts)

$$\int_0^1 \sqrt{9 + 4t^2} dt$$

- (c) Set up **but do not evaluate** an integral for the area of the surface produced by rotating the curve around the  $y$ -axis.

(5 pts)

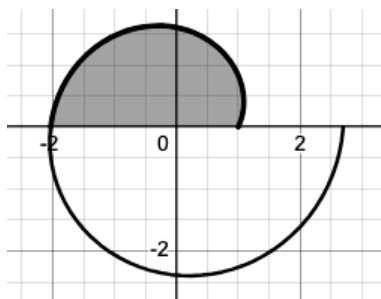
$$\begin{aligned} \text{radius} &= x \quad S.A. = 2\pi \int_0^1 x \, dS \\ &= 2\pi \int_0^1 3t \sqrt{9 + 4t^2} dt \end{aligned}$$

- (d) Set up **but do not evaluate** an integral for the area of the surface produced by rotating the curve around the  $x$ -axis.

(5 pts)

$$\begin{aligned} \text{rad} &= y = t^2 + 2 \\ S.A. &= 2\pi \int_0^1 (t^2 + 2) \sqrt{9 + 4t^2} dt \end{aligned}$$

6. Below is the graph of  $r = \sqrt{\theta - \frac{1}{\theta}}$ . Find the exact value of the shaded area. (10 pts)

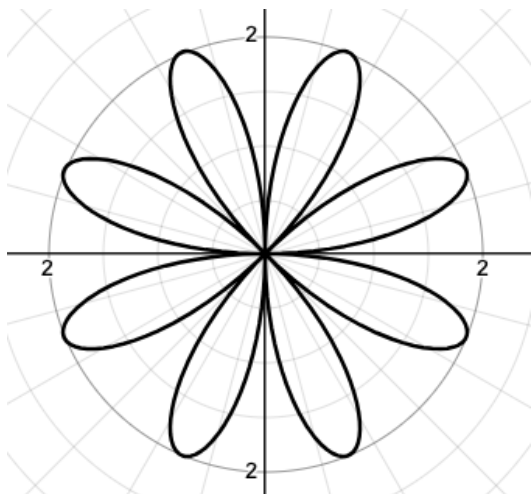


$$A = \frac{1}{2} \int_0^{\pi} (rad)^2 d\theta$$

7. Determine the number of petals on each of the following rose curves. (8 pts)

(a)  $r = 3 \sin(4\theta)$       (b)  $r = \frac{1}{2} \cos(10\theta)$       (c)  $r = 5 \sin(7\theta)$       (d)  $r = 3 \cos(5\theta)$

8. Set up, but don't evaluate an integral for the area of one petal of  $r = 2 \sin(4\theta)$  (8 pts)



1. For each of the following **sequences**, circle the correct answer. If it converges, tell the limit.

(1 pt each)

|                                                  |                      |                         |                         |
|--------------------------------------------------|----------------------|-------------------------|-------------------------|
| $a_n = (-1)^n \frac{3n^3 + 2n - 1}{6n^3 + 7}$    | Diverges to $\infty$ | Diverges by oscillating | Converges to _____      |
| $a_n = \frac{1 + \log n}{3n + 1}$                | Diverges to $\infty$ | Diverges by oscillating | Converges to <u>0</u>   |
| $a_n = \frac{2^n + n^2 + \ln n}{2n^3 + n^2 + 1}$ | Diverges to $\infty$ | Diverges by oscillating | Converges to _____      |
| $a_n = \frac{n^2 + 2n - 4}{3n^2 - 3n + 4}$       | Diverges to $\infty$ | Diverges by oscillating | Converges to <u>1/3</u> |

2. Suppose that  $a_n$  and  $b_n$  are **sequences**, and that  $\lim_{n \rightarrow \infty} a_n = \frac{2}{3}$  and  $\lim_{n \rightarrow \infty} b_n = 0$ . (1+1 pts)  
Circle the correct answer in each set of parentheses:

(a) The series  $\sum_{n=1}^{\infty} a_n$  ( Converges / Diverges / Not enough info to know )

(b) The series  $\sum_{n=1}^{\infty} b_n$  ( Converges / Diverges / Not enough info to know )

3. Find the sum of each geometric series

$$r = -\frac{2}{5}$$

(2+2 pts)

(a)  $\sum_{n=1}^{\infty} \frac{2^n}{5^{n+1}}$

(b)  $20 - 8 + \frac{16}{5} - \frac{32}{25} + \dots$

$$\frac{a}{1-r} = \frac{20}{1 + \frac{2}{5}}$$

$$= \frac{20}{\frac{7}{5}} = \frac{20 \cdot 5}{7} = \frac{100}{7}$$



4. Find the fourth partial sum  $s_4$  of this series.

(4 pts)

$$\sum_{n=0}^{\infty} \frac{n+1}{3^n} = \underbrace{\frac{1}{1} + \frac{2}{3} + \frac{3}{9} + \frac{4}{27}}_{1 + \frac{2}{3} + \frac{1}{3} + \frac{4}{27}} + \dots = \frac{58}{27}$$

5. Determine if each series Converges or Diverges. Write **C** or **D** in each blank.

(2 pts each)

C (a)  $\sum_{n=1}^{\infty} \frac{5^{n-1}}{7^n}$

D (b)  $\sum_{n=1}^{\infty} \frac{2^n}{n^3 + n + 1}$

C (c)  $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$

D (d)  $\sum_{n=1}^{\infty} \frac{3}{5n}$

C (e)  $\sum_{n=1}^{\infty} \frac{3}{n^5}$

D (f)  $\sum_{n=1}^{\infty} \frac{2^n + n}{2^n + n^2}$

C (g)  $\sum_{n=1}^{\infty} 4^{1-n} \sum \frac{4}{4^n}$

D (h)  $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$

C (i)  $\sum_{n=2}^{\infty} \frac{(-1)^{n+1}}{\ln n}$

6. Do these series converge or diverge? Briefly explain why.

(3+3 pts)

(a)  $\sum_{n=1}^{\infty} \left( \frac{1}{2^n} + \frac{1}{n^3} \right)$   
C C  
C

(b)  $\sum_{n=1}^{\infty} \left( \frac{1}{2^n} + \frac{3}{n} \right)$   
C D  
D

7. Use a comparison test to determine if each series converges or diverges. Tell what you're comparing to, but you don't need to show additional work. (3 pts each)

|                                                                                                                                 |                                                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><u>C</u> (a) <math>\sum_{n=1}^{\infty} \frac{n + \ln n}{n^3 + n^2 + n + 2}</math></p> <p><math>\sum \frac{1}{n^2}</math></p> | <p><u>C</u> (b) <math>\sum_{n=1}^{\infty} \frac{2 + \sin n}{1.001^n}</math></p> <p>C. <math>\sum \frac{3}{1.001^n}</math></p> <p><math>\sum \frac{2 + \sin n}{1.001^n} &lt; \sum \frac{3}{1.001^n}</math></p> |
| <p><u>D</u> (c) <math>\sum_{n=1}^{\infty} \frac{3^n + n^2}{2^n + n^{10}}</math></p> <p><math>\sum \frac{3^n}{2^n}</math></p>    | <p><u>D</u> (d) <math>\sum_{n=1}^{\infty} \frac{5 - 2n}{n^2 + \ln n + 1}</math></p> <p><math>\sum \frac{-2}{n} \quad D</math></p>                                                                             |

You must show your work to get credit (except on problem 1).

1. Each series below is convergent. Mark each as absolutely convergent (AC) or conditionally convergent (CC). You should be able to do these just by looking at them. (2 pts each)

(a)  $\sum_{n=1}^{\infty} \frac{(-1)^n}{3n+1}$  CC.  
 $\sum \frac{1}{3n+1}$  D

(b)  $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^3+n}$  AC  
 $\sum \frac{1}{n^3+n}$  C

(c)  $\sum_{n=1}^{\infty} \frac{2}{5^n}$  AC.

2. Use the ratio test to determine if the series converges or diverges. (6 pts)

$$\sum_{n=1}^{\infty} \frac{(-3)^n}{5^n n^4}$$

3. The series shown here has an interval of convergence with endpoints 3 and 5. Determine whether the endpoints should be included or excluded. Show your work and give your answer as an interval with the correct brackets and/or parentheses. (4 pts)

$$\sum_{n=0}^{\infty} \frac{(x-4)^n}{n}$$

$x=3$   
 $\sum \frac{(3-4)^n}{n} = \sum \frac{(-1)^n}{n}$  C

$x=5$   
 $\sum \frac{(5-4)^n}{n} = \sum \frac{1}{n}$  D

$\boxed{[3, 5)}$

4. Find the interval of convergence for the power series. Be sure to test the endpoints. (6 pts)

$$\sum_{n=0}^{\infty} \frac{(x-5)^n}{2^n}$$

- 
5. For the function  $f(x) = \frac{x^3}{1+3x^2}$

(a) Write out the first four nonzero terms of its series representation. (6 pts)

(b) Write out the first four nonzero terms of the series representation of  $f'(x)$ . (2 pts)

- 
6. For this function, write out the first four nonzero terms of its series representation. (6 pts)

$$g(x) = \frac{12}{3-x}$$

7. Express the geometric power series shown here using sigma notation. Combine and simplify your answer. (6 pts)

$$x^2 - \frac{1}{3}x^7 + \frac{1}{9}x^{12} - \frac{1}{27}x^{17} + \frac{1}{81}x^{22} + \dots$$

8. Write out the first four nonzero terms in the maclaurin series for each function: (4+4 pts)

(a)  $f(x) = \cos(x^2)$  :  $1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$

$$= 1 - \frac{(x^2)^2}{2!} + \frac{(x^2)^4}{4!} - \frac{(x^2)^6}{6!} + \dots = 1 - \frac{x^4}{2!} + \frac{x^8}{4!} - \frac{x^{12}}{6!} + \dots$$

(b)  $f(x) = x^3 e^x$   $\left( 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \right)$

$$= x^3 + \frac{x^4}{1!} + \frac{x^5}{2!} + \frac{x^6}{3!} + \dots$$

| Function | Maclaurin Series                                                                                                   | ROC          |
|----------|--------------------------------------------------------------------------------------------------------------------|--------------|
| $e^x$    | $= \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$                | $R = \infty$ |
| $\cos x$ | $= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$ | $R = \infty$ |